CHAPTER 1

Introduction to software project management

Objectives

When you have completed this chapter you will be able to:

- define the scope of 'software project management';
- understand some problems and concerns of software project managers;
- define the usual stages of a software project;
- explain the main elements of the role of management;
- appreciate the need for careful planning, monitoring and control;
- identify the stakeholders of a project and their objectives;
- define the success criteria for a project.

1.1 Introduction

This textbook is about '*software* project management'. The first question is whether the management of *software* projects is really that different from that of other projects. To answer this, we need to look at some key ideas about the planning, monitoring and control of software projects. We will see that all projects are about meeting objectives. Like any other project, a software project must satisfy real needs. To do this we must identify the project's stakeholders and their objectives. Ensuring that their objectives are met is the aim of project management. However, we cannot know that a project will meet its objectives in the future unless we know the present state of the project.

1.2 Why is software project management important?

This book is for students of software engineering and computer science and also those studying business information systems. More technically oriented students can be impatient at having to study something which keeps them away from their code. So why is it important to become familiar with project management?

The information in this paragraph comes from a National Audit Office report, Improving IT Procurement, November 2004.

First, there is the question of money. A lot of money is at stake with ICT projects. In the United Kingdom during the financial year 2002–2003, the central government spent more on contracts for ICT projects than on contracts related to roads (about £2.3 billion as opposed to £1.4 billion). The biggest departmental spender was the Department for Work and Pensions, who spent over £800 million on ICT. Mismanagement of ICT projects means that there is less to spend on good things such as hospitals.

There has been some debate about the precise validity of the Standish findings but the key point about the prevalence of IT project failings remains clear.

Unfortunately, projects are not always successful. In a report published in 2003, the Standish Group in the United States analysed 13,522 projects and concluded that only a third of projects were successful; 82% of projects were late and 43% exceeded their budget.

The reason for these project shortcomings is often the management of projects. The National Audit Office in the UK, for example, among other factors causing project failure identified '*lack of skills and proven approach to project management and risk management*'.

1.3 What is a project?

Dictionary definitions of 'project' include: 'A specific plan or design' 'A planned undertaking' 'A large undertaking: e.g. a public works scheme', *Longman Concise English Dictionary*, 1982.

The dictionary definitions put a clear emphasis on the project being a *planned* activity.

The emphasis on being planned assumes we can determine how to carry out a task before we start. Yet with exploratory projects this might be difficult. Planning is in essence thinking carefully about something before you do it – even with uncertain projects this is worth doing as long as the resulting plans are seen as provisional. Other activities, such as routine maintenance, will have been performed so many times that everyone knows exactly what to do. In these cases, planning hardly seems necessary, although procedures might be documented to ensure consistency and to help newcomers.

Programme management is often used to coordinate activities on concurrent jobs.

The activities that benefit most from conventional project management are likely to lie between these two extremes – see Figure 1.1.

There is a hazy boundary between the non-routine project and the routine job. The first time you do a routine task it will be like a project. On the other hand, a project to develop a system similar to previous ones that you have developed will have a large element of the routine.

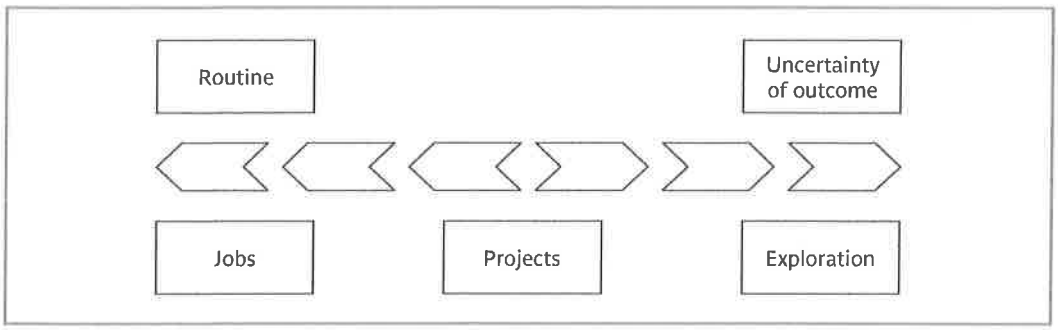


FIGURE 1.1 Activities most likely to benefit from project management

The following characteristics distinguish projects:

- non-routine tasks are involved;
- planning is required;
- specific objectives are to be met or a specified product is to be created;
- the project has a predetermined time span;
- work is carried out for someone other than yourself;
- work involves several specialisms;
- people are formed into a temporary work group to carry out the task;
- work is carried out in several phases;
- the resources that are available for use on the project are constrained;
- the project is large or complex.

The more any of these factors apply to a task, the more difficult that task will be. Project size is particularly important. The project that employs 20 developers is likely to be disproportionately more difficult than one with only 10 staff because of the need for additional coordination. The examples and exercises used in this book usually relate to smaller projects in order to make the techniques easier to grasp. However, the techniques and issues discussed are of equal relevance to larger projects.

EXERCISE 1.1

Consider the following:

- producing an edition of a newspaper;
- putting a robot vehicle on Mars to search for signs of life;
- getting married;
- amending a financial computer system to deal with a common European currency;

- a research project into what makes a good human–computer interface;
- an investigation into the reason why a user has a problem with a computer system;
- a second-year programming assignment for a computing student;
- writing an operating system for a new computer;
- installing a new version of a word processing package in an organization.

Some seem more like real projects than others. Put them into an order most closely matching your ideas of what constitutes a project. For each entry in the ordered list, describe the difference between it and the one above which makes it less worthy of the term 'project'.

There is no one correct answer to this exercise, but a possible solution to this and the other exercises you will come across may be found at the end of the book.

For example, see Rolf A. Lundin and Andres Söderholm (1995) 'A theory of the temporary organization' *Scandinavian Journal of Management* 11(4) 437–55.

Some argue that projects are especially problematic as they are temporary sub-organizations. A group of people is brought together to carry out a task. The existence of this sub-organization cuts across the authority of the existing units within the organization. This has the advantage that a group containing various specialists is focused on a single important task. However, the project is likely to be seen as disruptive to others. Also, expertise built up during the project may be lost when the team is eventually dispersed at the end of the project.

1.4 Software projects versus other types of project

F. P. Brooks (1987). 'No silver bullet: essence and accidents of software engineering'. This essay has been included in *The Mythical Man-Month*, Anniversary Edition, Addison Wesley, 1995.

Many techniques in general project management also apply to software project management, but Fred Brooks identified some characteristics of software projects which make them particularly difficult:

Invisibility When a physical artefact such as a bridge is constructed the progress can actually be seen. With software, progress is not immediately visible. Software project management can be seen as the process of making the invisible visible.

Complexity Per dollar, pound or euro spent, software products contain more complexity than other engineered artefacts.

Conformity The 'traditional' engineer usually works with physical systems and materials like cement and steel. These physical systems have complexity, but are governed by consistent physical laws. Software developers have to conform to the requirements of human clients. It is not just that individuals can be inconsistent. Organizations, because of lapses in collective memory, in internal communication or in effective decision making, can exhibit remarkable 'organizational stupidity'.

Flexibility That software is easy to change is seen as a strength. However, where the software system interfaces with a physical or organizational system, it is expected that the software will change to accommodate the other components rather than vice versa. Thus software systems are particularly subject to change.

1.5 Contract management and technical project management

In-house projects are where the users and the developers of new software work for the same organization. However, increasingly organizations contract out ICT development to outside developers. Here, the client organization will often appoint a 'project manager' to supervise the contract who will delegate many technically oriented decisions to the contractors. Thus, the project manager will not worry about estimating the effort needed to write individual software components as long as the overall project is within budget and on time. On the supplier side, there will need to be project managers who deal with the more technical issues. This book leans towards the concerns of these 'technical' project managers.

1.6 Activities covered by software project management

Chapter 4 on project analysis and technical planning looks at some alternative life cycles.

A software project is not only concerned with the actual writing of software. In fact, where a software application is bought in 'off the shelf', there may be no software writing as such, but this is still fundamentally a software project because so many of the other activities associated with software will still be present.

Usually there are three successive processes that bring a new system into being – see Figure 1.2.

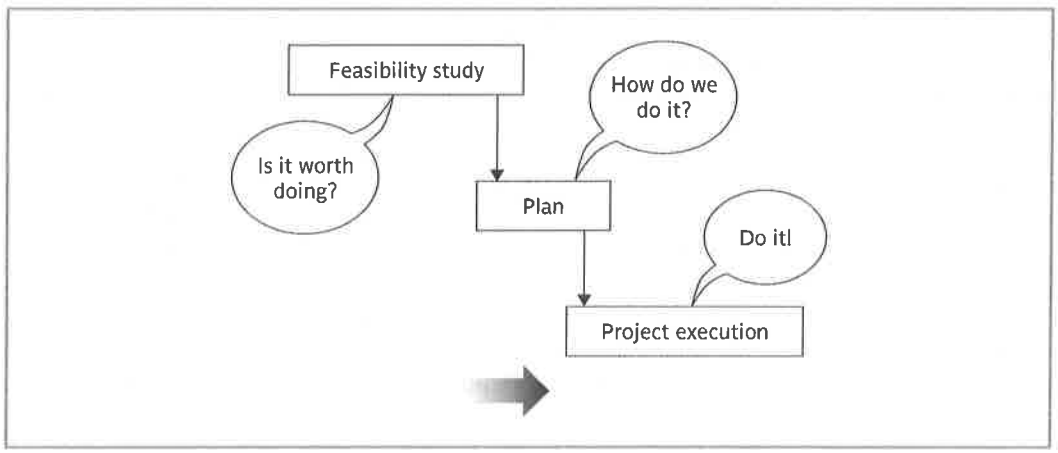


FIGURE 1.2 The feasibility study/plan/execution cycle

Chapter 2 explores some further aspects of programme management.

1 **The feasibility study** assesses whether a project is worth starting – that it has a valid *business case*. Information is gathered about the requirements of the proposed application. Requirements elicitation can, at least initially, be complex and difficult. The stakeholders may know the aims they wish to pursue, but not be sure about the means of achievement. The developmental and operational costs, and the value of the benefits of the new system, will also have to be estimated. With a large system, the feasibility study could be a project in its own right with its own plan. The study could be part of a strategic planning exercise examining a range of potential software developments. Sometimes an organization assesses a *programme* of development made up of a number of projects.

The PRINCE2 method, which is described in Appendix A, takes this iterative approach to planning. Annex 1 to this chapter has an outline of the content of a plan.

2 **Planning** If the feasibility study indicates that the prospective project appears viable, then project planning can start. For larger projects, we would not do all our detailed planning at the beginning. We create an outline plan for the whole project and a detailed one for the first stage. Because we will have more detailed and accurate project information after the earlier stages of the project have been completed, planning of the later stages is left to nearer their start.

3 **Project execution** The project can now be executed. The execution of a project often contains *design* and *implementation* sub-phases. Students new to project planning often find that the boundary between design and planning can be hazy. Design is making decisions about the form of the *products* to be created. This could relate to the external appearance of the software, that is, the user interface, or the internal architecture. The plan details the *activities* to be carried out to create these products. Planning and design can be confused because at the most detailed level, planning decisions are influenced by design decisions. Thus a software product with five major components is likely to require five sets of activities to create them.

Figure 1.3 suggests that these stages must be done strictly in sequence – we will see in Chapter 4 that other, iterative approaches can be adopted. However, the actual activities listed here would still be done.

Figure 1.3 shows the typical sequence of software development activities recommended in the international standard ISO 12207. Some activities are concerned with the *system* while others relate to *software*. The development of software will be only one part of a project. Software could be developed, for example, for a project which also requires the installation of an ICT infrastructure, the design of user jobs and user training.

- *Requirements analysis* starts with *requirements elicitation* or requirements gathering which establishes what the potential users and their managers require of the new system. It could relate to a *function* – that the system should do something. It could be a quality requirement – how well the functions must work. An example of this is dispatching an ambulance in response to an emergency telephone call. In this case transaction time would be affected by hardware and software performance as well as the speed of human operation. Training to ensure that operators use the computer system efficiently is an example of a *system requirement* for the project, as opposed to a specifically

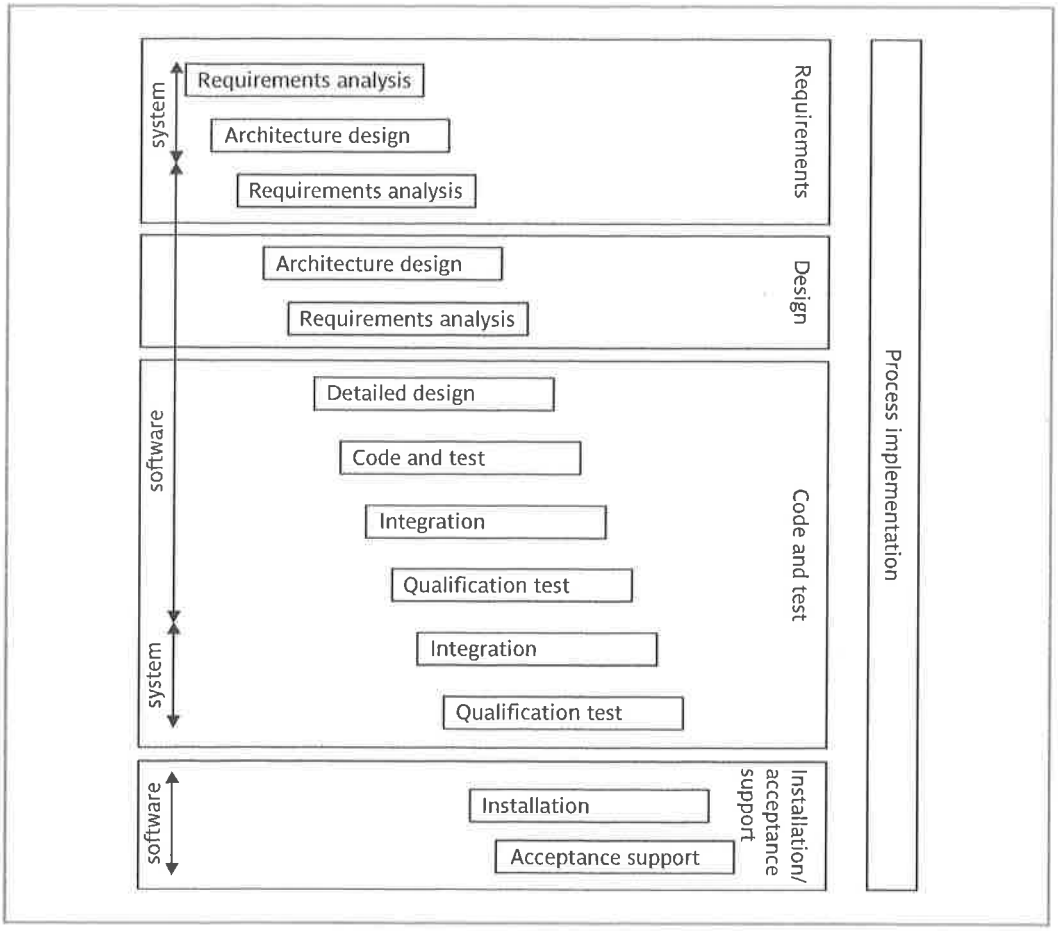


FIGURE 1.3 The ISO 12207 software development life cycle

software requirement. There would also be *resource requirements* that relate to application development costs.

- *Architecture design* The components of the new system that fulfil each requirement have to be identified. Existing components may be able to satisfy some requirements. In other cases, a new component will have to be made. These components are not only software: they could be new hardware or work processes. Although software developers are primarily concerned with software components, it is very rare that these can be developed in isolation. They will, for example, have to take account of existing legacy systems with which they will interoperate. The design of the *system architecture* is thus an input to the *software requirements*. A second architecture design process then takes place that maps the software requirements to *software components*.
- *Detailed design* Each software component is made up of a number of software units that can be separately coded and tested. The detailed design of these units is carried out separately.

- *Code and test* refers to writing code for each software unit. Initial testing to debug individual software units would be carried out at this stage.
- *Integration* The components are tested together to see if they meet the overall requirements. Integration could involve combining different software components, or combining and testing the software element of the system in conjunction with the hardware platforms and user interactions.
- *Qualification testing* The system, including the software components, has to be tested carefully to ensure that all the requirements have been fulfilled.
- *Installation* This is the process of making the new system operational. It would include activities such as setting up standing data (for example, the details for employees in a payroll system), setting system parameters, installing the software onto the hardware platforms and user training.
- *Acceptance support* This is the resolving of problems with the newly installed system, including the correction of any errors, and implementing agreed extensions and improvements. Software maintenance can be seen as a series of minor software projects. In many environments, most software development is in fact maintenance.

EXERCISE 1.2

Brightmouth College is a higher education institution which used to be managed by a local government authority but has now become autonomous. Its payroll is still administered by the local authority and pay slips and other output are produced in the local authority's computer centre. The authority now charges the college for this service. The college management are of the opinion that it would be cheaper to obtain an 'off-the-shelf' payroll package and do the payroll processing themselves.

What would be the main stages of the project to convert to independent payroll processing by the college? Bearing in mind that an off-the-shelf package is to be used, how would this project differ from one where the software was to be written from scratch?

1.7 Plans, methods and methodologies

A plan for an activity must be based on some idea of a *method* of work. For example, if you were asked to test some software, you may know nothing about the software to be tested, but you could assume that you would need to:

- analyse the requirements for the software;
- devise and write test cases that will check that each requirement has been satisfied;
- create test scripts and expected results for each test case;
- compare the actual results and the expected results and identify discrepancies.

While a *method* relates to a type of activity in general, a *plan* takes that method (and perhaps others) and converts it to real activities, identifying for each activity:

- its start and end dates;
- who will carry it out;
- what tools and materials – including information – will be needed.

The output from one method might be the input to another. Groups of methods or techniques are often grouped into *methodologies* such as object-oriented design.

EXERCISE 1.3

This should ideally be done in groups of about four, but you can think about how you would go about this exercise on your own if needs be. You are probably in a building that has more than one storey. From the point of view of this exercise, the bigger the building the better.

In a group of four, work out how you would obtain an accurate estimate of the height of the building. (If you happen to be in a single-storey building, you can estimate the floor area instead!) Plan how you would carry out any actions needed to obtain your estimate. Spend 20 minutes on this – you must remain in the same room for this planning phase. Once planning is complete, implement your plan, timing how long it takes to produce your final figure.

If there is more than one group carrying out this exercise, after completion of the task you can compare answers and also the approach you used when coming up with your answer.

1.8 Some ways of categorizing software projects

Projects may differ because of the different technical products to be created. Thus we need to identify the characteristics of a project which could affect the way in which it should be planned and managed. Other factors are discussed below.

Compulsory versus voluntary users

In workplaces there are systems that staff have to use if they want to do something, such as recording a sale. However, use of a system is increasingly voluntary, as in the case of computer games. Here it is difficult to elicit precise requirements from potential users as we could with a business system. What the game will do will thus depend much on the informed ingenuity of the developers, along with techniques such as market surveys, focus groups and prototype evaluation.

Information systems versus embedded systems

Embedded systems are also called real-time or industrial systems.

A traditional distinction has been between *information systems* which enable staff to carry out office processes and *embedded systems* which control machines. A stock control system would be an information system. An embedded, or process control, system might control the air conditioning equipment in a building. Some systems may have elements of both where, for example, the stock control system also controls an automated warehouse.

EXERCISE 1.4

Would an operating system on a computer be an information system or an embedded system?

Objectives versus products

Projects may be distinguished by whether their aim is to produce a *product* or to meet certain *objectives*.

A project might be to create a product, the details of which have been specified by the client. The client has the responsibility for justifying the product.

Service level agreements are becoming increasingly important as organizations contract out functions to external service suppliers.

On the other hand, the project requirement might be to meet certain objectives which could be met in a number of ways. An organization might have a problem and ask a specialist to recommend a solution.

Many software projects have two stages. First is an objective-driven project resulting in recommendations. This might identify the need for a new software system. The next stage is a project actually to create the software product.

This is useful where the technical work is being done by an external group and the user needs are unclear at the outset. The external group can produce a preliminary design at a fixed fee. If the design is acceptable the developers can then quote a price for the second, implementation, stage based on an agreed requirement.

EXERCISE 1.5

Would the project to implement an independent payroll system at the Brightmouth College described in Exercise 1.2 above be an objective-driven project or a product-driven project?

1.9 Stakeholders

These are people who have a stake or interest in the project. Their early identification is important as you need to set up adequate communication channels with them. Stakeholders can be categorized as:

- *Internal to the project team* This means that they will be under the direct managerial control of the project leader.
- *External to the project team but within the same organization* For example, the project leader might need the assistance of the users to carry out systems testing. Here the commitment of the people involved has to be negotiated.
- *External to both the project team and the organization* External stakeholders may be customers (or users) who will benefit from the system that the project implements. They may be contractors who will carry out work for the project. The relationship here is usually based on a contract.

B. W. Boehm and R. Ross, 'Theory W software project management: principles and examples', in B. W. Boehm (ed.) (1989) *Software Risk Management*, IEEE Computer Society Press.

Different types of stakeholder may have different objectives and one of the jobs of the project leader is to recognize these different interests and to be able to reconcile them. For example, end-users may be concerned with the ease of use of the new application, while their managers may be more focused on staff savings. The project leader therefore needs to be a good communicator and negotiator. Boehm and Ross proposed a 'Theory W' of software project management where the manager concentrates on creating situations where all parties benefit from a project and therefore have an interest in its success. (The 'W' stands for 'win–win'.)

Project managers can sometimes miss an important stakeholder group, especially in unfamiliar business contexts. These could be departments supplying important services that are taken for granted.

Given the importance of coordinating the efforts of stakeholders, the recommended practice is for a *communication plan* to be created at the start of a project.

The role and format of communication plans will be explained in greater detail in Chapter 11 on managing people in software environments.

EXERCISE 1.6

Identify the stakeholders in the Brightmouth College payroll project.

1.10 Setting objectives

Among all these stakeholders are those who actually own the project. They control the financing of the project. They also set the objectives of the project. The objectives should define what the project team must achieve for project success. Although different

stakeholders have different motivations, the project objectives identify the shared intentions for the project.

Objectives focus on the desired outcomes of the project rather than the tasks within it – they are the 'post-conditions' of the project. Informally the objectives could be written as a set of statements following the opening words *'the project will be a success if. . . .'* Thus one statement in a set of objectives might be *'customers can order our products online'* rather than *'to build an e-commerce website'*. There is often more than one way to meet an objective and the more possible routes to success the better.

There may be several stakeholders, including users in different business areas, who might have some claim to project ownership. In such a case, a *project authority* needs to be explicitly identified with overall authority over the project.

This committee is likely to contain user, development and management representatives.

This authority is often a *project steering committee* (or *project board* or *project management board*) with overall responsibility for setting, monitoring and modifying objectives. The project manager runs the project on a day-to-day basis, but regularly reports to the steering committee.

Sub-objectives and goals

Defining sub-objectives requires assumptions about how the main objective is to be achieved.

An effective objective for an individual must be something that is within the control of that individual. An objective might be that the software application produced must pay for itself by reducing staff costs. As an overall business objective this might be reasonable. For software developers it would be unreasonable as any reduction in operational staff costs depends not just on them but on the operational management of the delivered system. A more appropriate *goal* or sub-objective for the software developers would be to keep development costs within a certain budget.

We can say that in order to achieve the objective we must achieve certain goals or sub-objectives first. These are steps on the way to achieving an objective, just as goals scored in a football match are steps towards the objective of winning the match. Informally this can be expressed as a set of statements following the words *'To reach objective . . . , the following must be in place . . .'*.

The mnemonic SMART is sometimes used to describe well-defined objectives:

- *Specific* Effective objectives are concrete and well defined. Vague aspirations such as *'to improve customer relations'* are unsatisfactory. Objectives should be defined so that it is obvious to all whether the project has been successful.
- *Measurable* Ideally there should be *measures of effectiveness* which tell us how successful the project has been. For example, *'to reduce customer complaints'* would be more satisfactory as an objective than *'to improve customer relations'*. The measure can, in some cases, be an answer to simple yes/no question, e.g. *'Did we install the new software by 1 June?'*

This still leaves a problem about the level at which the target should be set, e.g. why, say, a 50% reduction in complaints and not 40% or 60%?

- *Achievable* It must be within the power of the individual or group to achieve the objective.
- *Relevant* The objective must be relevant to the true purpose of the project.

- *Time constrained* There should be a defined point in time by which the objective should have been achieved.

EXERCISE 1.7

Bearing in mind the above discussion of objectives, comment on the appropriateness of the wording of each of the following 'objectives' for software developers:

(i) to implement the new application on time and within budget;

(ii) to implement the new software application with the fewest possible software errors that might lead to operational failures;

(iii) to design a system that is user-friendly;

(iv) to produce full documentation for the new system.

Measures of effectiveness

These concepts are explained more fully in Chapter 13 on software quality.

Measures of effectiveness provide practical methods of checking that an objective has been met. 'Mean time between failures' (mtbf) might, for example, be used to measure reliability. This is a *performance* measurement and, as such, can only be taken once the system is operational. Project managers want to get some idea of the performance of the completed system as it is being constructed. They will therefore seek *predictive* measures. For example, a large number of errors found during code inspections might indicate potential problems with reliability later.

EXERCISE 1.8

Identify the objectives and sub-objectives of the Brightmouth College payroll project. What measures of effectiveness could be used to check the success in achieving the objectives of the project?

1.11 The business case

The business case should be established at the time of the project's feasibility study. Chapter 2 explains the idea of a business case in more detail.

Most projects need to have a justification or business case: the effort and expense of pushing the project through must be seen to be worthwhile in terms of the benefits that will eventually be felt. A cost–benefit analysis will often be part of the project's feasibility study. This will itemize and quantify the project's costs and benefits. The benefits will be affected by the completion date: the sooner the project is completed, the sooner the benefits can be experienced. The quantification of benefits will often require the formulation of a *business model* which explains how the new application can generate the claimed benefits.

A simple example of a business model is that a new web-based application might allow customers from all over the world to order a firm's products via the internet, increasing sales and thus increasing revenue and profits.

Any project plan must ensure that the business case is kept intact. For example:

- that development costs are not allowed to rise to a level which threatens to exceed the value of benefits;
- that the features of the system are not reduced to a level where the expected benefits cannot be realized;
- that the delivery date is not delayed so that there is an unacceptable loss of benefits.

1.12 Project success and failure

A good introduction to the issues discussed here can be found in A. J. Shenhar and O. Levy (1997) 'Mapping the dimensions of project success' *Project Management Journal* 28(2) 9–12.

The project plan should be designed to ensure project success by preserving the business case for the project. However, every non-trivial project will have problems, and at what stage do we say that a project is actually a failure? Because different stakeholders have different interests, some stakeholders in a project might see it as a success while others do not.

Broadly speaking, we can distinguish between *project objectives* and *business objectives*. The project objectives are the targets that the project team is expected to achieve. In the case of software projects, they can usually be summarized as delivering:

- the agreed functionality
- to the required level of quality
- on time
- within budget.

A project could meet these targets but the application, once delivered could fail to meet the business case. A computer game could be delivered on time and within budget, but might then not sell. A commercial website used for online sales could be created successfully, but customers might not use it to buy products, because they could buy the goods more cheaply elsewhere.

The assessment of the value of project benefits is explored in greater depth in Chapter 2.

We have seen that in business terms it can generally be said that a project is a success if the value of benefits exceeds the costs. We have also seen that while project managers have considerable control over development costs, the value of the benefits of the project deliverables is dependent on external factors such as the number of customers. Project objectives still have some bearing on eventual business success. As we will see in Chapter 2, increasing development costs reduce the chances of the delivered product being profitable. A delay in completion reduces the amount of time during which benefits can be generated and diminishes the value of the project.

A project can be a success on delivery but then be a business failure, On the other hand, a project could be late and over budget, but its deliverables could still, over time, generate benefits that outweigh the initial expenditure.

Some argue that the possible gap between project and business concerns can be reduced by having a broader view of projects that includes business issues. For example, the project management of an e-commerce website implementation could plan activities such as market surveys, competitor analysis, focus groups, prototyping, and evaluation by typical potential users – all designed to reduce business risks.

For a wider discussion of the relationships between successive projects, see M. Engwall (2003) 'No project is an island: linking projects to history and context' *Research Policy* 32 789–808.

Because the focus of project management is, not unnaturally, on the immediate project, it may not be seen that the project is actually one of a sequence. Later projects benefit from the technical skills learnt on earlier projects. Technical learning will increase costs on the earlier projects, but later projects benefit as the learnt technologies can be deployed more quickly, cheaply and accurately. This expertise is often accompanied by additional software assets, for example reusable code. Where software development is outsourced, there may be immediate savings, but these longer-term benefits of increased expertise will be lost. Astute managers may assess which areas of technical expertise it would be beneficial to develop.

Customer relationships can also be built up over a number of projects. If a client has trust in a supplier who has done satisfactory work in the past, they are more likely to use that company again, particularly if the new requirement builds on functionality already delivered. It is much more expensive to acquire new clients than it is to retain existing ones.

1.13 What is management?

We have explored some of the special characteristics of software. We now look at the 'management' aspect of software project management. It has been suggested that management involves the following activities:

- planning – deciding what is to be done;
- organizing – making arrangements;
- staffing – selecting the right people for the job etc.;
- directing – giving instructions;
- monitoring – checking on progress;
- controlling – taking action to remedy hold-ups;
- innovating – coming up with new solutions;
- representing – liaising with clients, users, developer, suppliers and other stakeholders.

EXERCISE 1.9

Paul Duggan is the manager of a software development section. On Tuesday at 10.00 a.m. he and his fellow section heads have a meeting with their group manager about the staffing requirements for the coming year. Paul has already drafted a document 'bidding' for staff. This is based on the work planned for his section for the next year. The document is discussed at the meeting. At 2.00 p.m. Paul has a meeting with his senior staff about an important project his section is undertaking. One of the programming staff has just had a road accident and will be in hospital for some time. It is decided that the project can be kept on schedule by transferring another team member from less urgent work to this project. A temporary replacement is to be brought in to do the less urgent work but this may take a week or so to arrange. Paul has to phone both the human resources manager about getting a replacement and the user for whom the less urgent work is being done, explaining why it is likely to be delayed.

Identify which of the eight management responsibilities listed above Paul was responding to at different points during his day.

1.14 Management control

Management, in general, involves setting objectives for a system and then monitoring the performance of the system. In Figure 1.4 the 'real world' is shown as being rather formless. Especially in the case of large undertakings, there will be a lot going on about which management should be aware.

EXERCISE 1.10

An ICT project is to replace locally held paper-based records with a centrally organized database. Staff in a large number of offices that are geographically dispersed need training and will then have to use the new ICT system to set up the backlog of manual records on the new database. The system cannot be properly operational until the last record has been transferred. The new system will only be successful if new transactions can be processed within certain time cycles.

Identify the data that you would collect to ensure that during execution of the project things were going to plan.

This will involve the local managers in *data collection*. Bare details, such as 'location X has processed 2000 documents', will not be very useful to higher management: *data processing* will be needed to transform this raw *data* into useful *information*. This might be in such forms as 'percentage of records processed', 'average documents processed per day per person' and 'estimated completion date'.

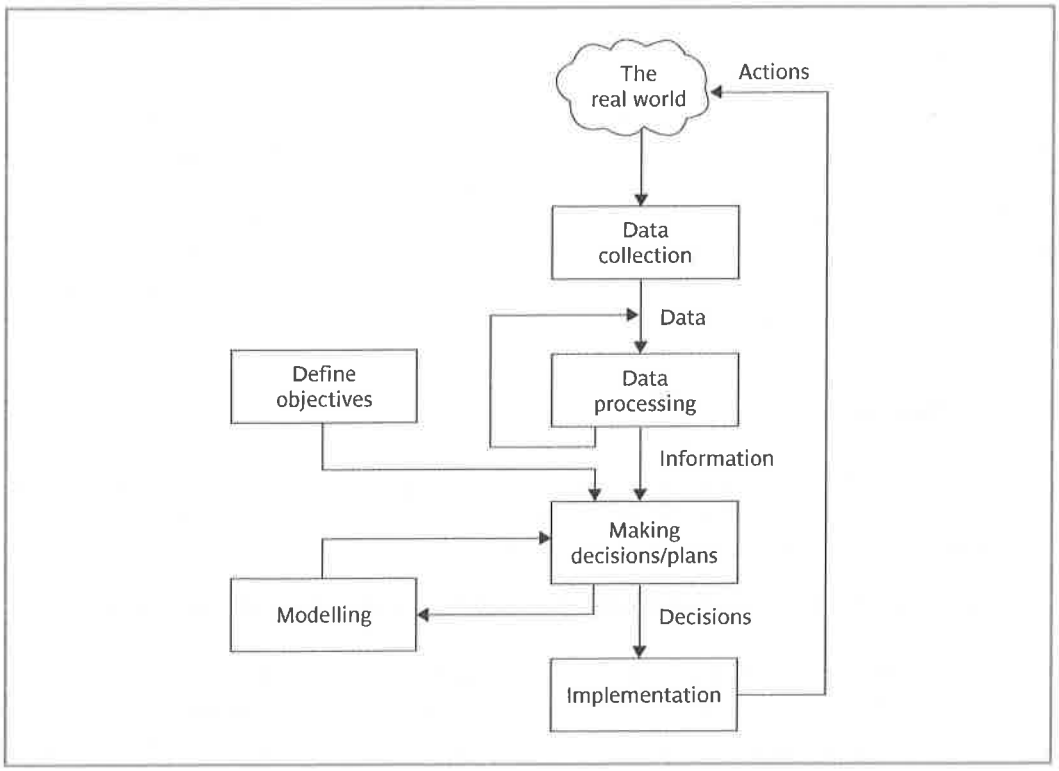


FIGURE 1.4 The project control cycle

In our example, the project management might examine the 'estimated completion date' for completing data transfer for each branch. These can be checked against the overall target date for completion of this phase of the project. In effect they are comparing actual performance with one aspect of the overall project objectives. They might find that one or two branches will fail to complete the transfer of details in time. They would then need to consider what to do (this is represented in Figure 1.4 by the box *Making decisions/plans*). One possibility would be to move staff temporarily from one branch to another. If this is done, there is always the danger that while the completion date for the one branch is pulled back to before the overall target date, the date for the branch from which staff are being moved is pushed forward beyond that date. The project manager would need to calculate carefully what the impact would be in moving staff from particular branches. This is *modelling* the consequences of a potential solution. Several different proposals could be modelled in this way before one was chosen for *implementation.*

Having implemented the decision, the situation needs to be kept under review by collecting and processing further progress details. For instance, the next time that progress is reported, a branch to which staff have been transferred could still be behind in transferring details. This might be because the reason why the branch has got behind in transferring details is because the manual records are incomplete and another department,

for whom the project has a low priority, has to be involved in providing the missing information. In this case, transferring extra staff to do data inputting will not have accelerated data transfer.

It can be seen that a project plan is dynamic and will need constant adjustment during the execution of the project. Courses and books on project management (such as this one) often focus considerable attention on project planning. While this is to be expected, with nearly all projects much more time is spent actually doing the project rather than planning it. A good plan provides a foundation for a good project, but is nothing without intelligent execution. The original plan will not be set in stone but will be modified to take account of changing circumstances.

1.15 Conclusion

This chapter has laid a foundation for the remainder of the book by defining what is meant by various terms such as 'software project' and 'management'. Among some of the more important points that have been made are the following:

- Projects are by definition non-routine and therefore more uncertain than normal undertakings.
- Software projects are similar to other projects but have some attributes that present particular difficulties, e.g. the relative invisibility of many of their products.
- A key factor in project success is having clear objectives. Different stakeholders in a project, however, are likely to have different objectives. This points to the need for a recognized overall project authority.
- For objectives to be effective there must be practical ways of testing that the objectives have been met.
- Where projects involve many different people, effective channels of information have to be established. Having objective measures of success helps unambiguous communication between the various parties to a project.

Annex 1 Contents list for a project plan

The detail that goes into these sections will be explained in later chapters. For example, Chapter 7 relates to risk while Chapter 13 explains aspects of the management of quality.

- Introduction
- Background: including reference to the business case
- Project objectives
- Constraints – these could be included with project objectives
- Methods
- Project products: both deliverable products that the client will receive and intermediate products
- Activities to be carried out
- Resources to be used

- Risks to the project
- Management of the project, including
  - organizational responsibilities
  - management of quality
  - configuration management

1.16 Further exercises

1 List the problems you experienced when you carried out a recent ICT-related assignment. Try to put these problems into some order of magnitude. For each problem consider whether there was some way in which the problem could have been reduced by better organization and planning by yourself.

2 Identify the main types of personnel employed in an information systems department. For each stage of a typical IS development project, list the types of personnel who are likely to be involved.

3 A public library is considering the implementation of a computer-based system to help administer book loans at libraries. Identify the stakeholders in such a project. What might be the objectives of such a project and how might the success of the project be measured in practical terms?

4 A software house has developed a customized order processing system for a client. You are an employee of the software house that has been asked to organize a training course for the end-users of the system. At present, a user handbook has been produced, but no specific training material. A plan is now needed for the project which will set up the delivery of the training courses. The project can be assumed to have been completed when the first training course starts. Among the things that will need to be considered are the following:

   - training materials will need to be designed and created;
   - a timetable will need to be drafted and agreed;
   - date(s) for the course will need to be arranged;
   - the people attending the course will need to be identified and notified;
   - rooms and computer facilities for the course will need to be provided.

   A Identify the main stakeholders for this project.
   B Draw up objectives for this project.
   C For the objectives, identify the measures of effectiveness.
   D For each objective, write down sub-objectives or goals and the stakeholders who will be responsible for their achievement.

5 A manager is in charge of a sub-project of a larger project. The sub-project requires the transfer of paper documents into a computer-based document retrieval system and their subsequent indexing so that they can be accessed via key-words. Optical character readers are to be used for the initial transfer but the text then needs to be clerically checked and corrected by staff. The project is currently scheduled to take 12 months using permanent staff. A small budget is available to hire temporary staff in the case of

staff absences through holidays, sickness or temporary transfer to other, more urgent, jobs. Discuss the control system that will need to be in place to control that sub-project.

6 The idea behind a project is that students should be able to access details of available placements via an intranet. When there is a placement opportunity for which they wish to be considered, they would be able to apply for it electronically. This would cause a copy of their CV, which would also be held online, to be sent to the potential employer.

Details of interviews and placement offers would all be sent by e-mail. While some human intervention would be needed, the process would be automated as far as possible.

You are required to produce a business case report for such an application, which justifies the potential development by showing that the value of its potential benefits outweighs its development and operational costs.

Create lists of the main benefits and costs for the project. You do not have to specify actual figures, just the headings under which they would appear.